

Number Algorithms and Patterns

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Name: _____

Date: _____

Learning Intention

I can follow and create step-by-step number algorithms, make decisions and describe patterns that emerge.

Strategy Box: an algorithm is a clear sequence of steps

Example rule: Start at 4. Add 3. Repeat 4 times.

4 → 7 → 10 → 13 → 16

Pattern noticed: every number increases by 3.

Worked Example - Follow the Steps

Start at 6 → multiply by 2 → add 4 → subtract 3

6 → 12 → 16 → 13

A. Warm Up - Follow Each Algorithm

1. Start at 5. Add 4 three times:

5 → ____ → ____ → ____

2. Start at 30. Subtract 5 four times:

30 → ____ → ____ → ____ → ____

3. Start at 3. Multiply by 4, then add 6:

3 → ____ → ____

Picture Practice - Number Path

P1. Follow: Start at 8. Add 5 each time. Complete the path and describe the pattern.



Pattern: _____

Quick check: Follow every step in order and record each result.

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B. Build Your Skills - Make Decisions

Use this rule: If the number is even, divide by 2. If it is odd, add 5.

4. Start with 18. It is _____, so _____. Result: _____

5. Start with 13. It is _____, so _____. Result: _____

6. Start with 26. Apply the rule twice: $26 \rightarrow _____ \rightarrow _____$

7. Start with 7. Apply the rule three times: $7 \rightarrow _____ \rightarrow _____ \rightarrow _____$

C. Create and Investigate

8. Create an algorithm with 3 steps that begins at 5 and ends at 25.

Step 1: _____ Step 2: _____ Step 3: _____

Number path: $5 \rightarrow _____ \rightarrow _____ \rightarrow 25$

9. Rule: Start at 2, multiply by 2, then add 1. Repeat using each new result.

$2 \rightarrow _____ \rightarrow _____ \rightarrow _____$

Describe the emerging pattern: _____

D. Apply It - Explain Your Thinking

10. Ava followed "start at 10, add 4 three times" and wrote 10, 14, 18, 24.

Find her error and write the correct sequence.

11. Why must algorithm steps be followed in the correct order?

Number Algorithms and Patterns - Answer Key

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Learning check

Check that students follow steps in order, apply decisions correctly and describe a pattern using precise mathematical language.

A. Warm Up and Picture Practice

1. $5 \rightarrow 9 \rightarrow 13 \rightarrow 17$.

2. $30 \rightarrow 25 \rightarrow 20 \rightarrow 15 \rightarrow 10$.

3. $3 \rightarrow 12 \rightarrow 18$.

P1. $8 \rightarrow 13 \rightarrow 18 \rightarrow 23 \rightarrow 28$; the numbers increase by 5.

B. Make Decisions

4. Even; divide by 2; result 9.

5. Odd; add 5; result 18.

6. $26 \rightarrow 13 \rightarrow 18$.

7. $7 \rightarrow 12 \rightarrow 6 \rightarrow 3$.

C-D. Create, Investigate and Explain

8. Answers vary. Example: $+5, \times 2, +5$ gives $5 \rightarrow 10 \rightarrow 20 \rightarrow 25$.

9. $2 \rightarrow 5 \rightarrow 11 \rightarrow 23$. Each result is double the previous number, plus 1.

10. Ava added 6 in the last step. Correct sequence: 10, 14, 18, 22.

11. Sample: Changing the order can change the result; each step uses the previous result.

Teacher feedback checklist

- ☐ Follows steps in order
- ☐ Records each result
- ☐ Applies decisions correctly
- ☐ Describes the pattern clearly
- ☐ Creates a valid algorithm